

Java Code

```
1  import java.util.ArrayList;
2  import java.util.List;
3  import java.util.Scanner;
4
5  public class StudentGrading {
6
7      public static void main(String[] args) {
8          Scanner scanner = new Scanner(System.in);
9
10         System.out.println("=====");
11         System.out.println("          STUDENT GRADING SYSTEM          ");
12         System.out.println("=====");
13         System.out.println();
14
15         // Input number of students (n)
16         System.out.print("Number of students : ");
17         int n = Integer.parseInt(scanner.nextLine().trim());
18
19         // Input student names and their scores
20         String[] names = new String[n];
21         double[] scores = new double[n];
22
23         for (int j = 0; j < n; j++) {
24             System.out.println();
25             System.out.println("-----");
26             System.out.println(" Student " + (j + 1) + " of " + n);
27             System.out.println("-----");
28             System.out.print(" Name : ");
29             names[j] = scanner.nextLine().trim();
30
31             System.out.print(" Score : ");
32             scores[j] = Double.parseDouble(scanner.nextLine().trim());
33         }
34
35         System.out.println();
36         System.out.println("=====");
37         System.out.println("          PROCESSING RESULTS...          ");
38         System.out.println("=====");
39
40         // Initialize variables
41         double sum = 0;
42         int passCount = 0;
43         int failCount = 0;
44         double highestScore = -1;
45         List<String> topStudents = new ArrayList<>();
46         String[] grades = new String[n]; // parallel array to store letter grades
47
48         int i = 0;
49         while (i < n) {
50             String currentName = names[i];
```

```
51         double currentScore = scores[i];
52
53         // Determine letter grade based on grading scale
54         String grade;
55         if (currentScore >= 98) {
56             grade = "A+";
57         } else if (currentScore >= 92) {
58             grade = "A";
59         } else if (currentScore >= 87) {
60             grade = "B+";
61         } else if (currentScore >= 81) {
62             grade = "B";
63         } else if (currentScore >= 77) {
64             grade = "C+";
65         } else if (currentScore >= 71) {
66             grade = "C";
67         } else if (currentScore >= 60) {
68             grade = "D";
69         } else {
70             grade = "F";
71         }
72
73         grades[i] = grade; // Store letter grade for this student
74
75         // Track top-performing student(s)
76         if (currentScore > highestScore) {
77             highestScore = currentScore;
78             topStudents.clear();
79             topStudents.add(currentName);
80         } else if (currentScore == highestScore) {
81             topStudents.add(currentName);
82         }
83         // else: do nothing
84
85         // Update sum and pass/fail counts
86         sum = sum + currentScore;
87         if (currentScore >= 60) {
88             passCount = passCount + 1;
89         } else {
90             failCount = failCount + 1;
91         }
92
93         i = i + 1;
94     }
95
96     double classAverage = sum / n;
97
98     boolean below70;
99     if (classAverage < 70) {
100         below70 = true;
101     } else {
102         below70 = false;
103     }
104 }
```

```
105         // Output
106         System.out.println();
107         System.out.println("=== Letter Grades ===");
108         for (int k = 0; k < n; k++) {
109             System.out.println(names[k] + " - Score: " + scores[k] + " - Grade: " +
grades[k]);
110         }
111
112         System.out.println();
113         System.out.println("=== Top-Performing Student(s) ===");
114         System.out.println("Score: " + highestScore);
115         for (String topStudent : topStudents) {
116             System.out.println(topStudent);
117         }
118
119         System.out.println();
120         System.out.printf("Class Average: %.2f%n", classAverage);
121         System.out.println("Number of students who passed (score >= 60): " +
passCount);
122         System.out.println("Number of students who did not pass (score < 60): " +
failCount);
123         System.out.println("Is class average below 70? " + below70);
124
125         scanner.close();
126     }
127 }
128
```